

COMP101

Introduction to Programming

Tutorial-06

Pseudocode Iteration

Iteration (for loop, while loop)

- Constructs that repeat ('iterate') the flow of control of a program
- Pseudocode is not language-specific
 - Python uses 'while loops' and 'for loops'
 - we only use 'loop' and 'loopend' in pseudocode language sub-set
 - implementation of for or while is left to when we program it
- We use pseudocode to identify the logic of iteration

Problem statement

- A die (singular of the plural 'dice') is a six-sided cube with each side of the cube showing an integer 1 to 6. There is one unique integer on each side.
- A user 'throws' two die. If they throw a 'double six' they are told they can play the game again. This continues until a number other than 12 is thrown and play is passed to the next player.

Draft Solution

1. input

1.1 input integer1

1.2 input integer2

2. process

2.1 calc total = integer1 + integer2

2.2 loop

2.2.1 if total = 12 then

2.2.1.1 print 'Throw again'

2.2.1.1 loop

2.2.1.1.1 input integer1

2.2.1.1.2 input integer2

2.2.3 loopend

2.2.2 else

2.2.2.1 print 'pass dice'

2.3 loopend

3. Output

3.1 print game ended

We are not trying to implement the program at this stage but to establish its logic ready for coding.

For instance, here we would probably code an import random function – but this is language specific and has no place here.

No initialization – trivial design

Actual loop type not defined at pseudocode – implementation will determine for loop or while loop

'loopend' not compulsory but helps identify the logic of where inner loop ends and outer loop ends

Nested loop